

UX Design Studio — Source of Truth

Module: UX Design Studio — Visual Preview & Approval Workbench for InsaneSDD 2.0 **Type:** Proof-of-work concept extension (not an official InfoBeans product) **Author:** [Your name] · **Target date:** July 17, 2026 · **Budget:** ≤ 50 build hours **Version:** 1.0 (frozen — scope changes require a new version of this document)

1. Purpose

InsaneSDD 2.0's differentiator is human-supervised AI delivery: approval gates at requirement interpretation, baseline freezing, UX design, and execution. The UX gate is the weakest of the four — the UX Design Agent outputs personas, journeys, screen lists, a component inventory, and a design system **as text**, and asks stakeholders to click "Approve UX Design" without ever seeing a rendered screen. The first time anyone sees the actual UI is after code is generated, branches are created, and PRs are merged — exactly the late-discovery rework InsaneSDD exists to eliminate.

UX Design Studio closes this gap. It renders the AI-generated UX specification as **interactive, navigable, design-token-themed wireframes**, and replaces the all-or-nothing gate with **per-screen approval and structured revision requests** that remain traceable to the frozen requirement baseline.

Purpose of the POC specifically: demonstrate lead-frontend engineering (spec-driven rendering engine, runtime theming, component registry, state architecture) and product ownership (gap identification validated against InfoBeans' own sources) in a form InfoBeans could plausibly adopt.

2. Target users and buyers

Role	Relationship to module	Grounding
Product Owner / PM ("Alex" in the AgentPilot demo)	Primary user — reviews and approves screens before execution	Demo persona
UX/Design stakeholders	Validate design-system application, journeys, responsive behavior	UX Design module in demo
Compliance Officer ("Taylor")	Verifies accessibility annotations before sign-off	Demo persona; WCAG 2.1 AA is in the generated spec
Frontend/Engineering Lead ("Jordan"-adjacent)	Confirms component inventory and screen structure match what agents will build	Demo persona
Buyer: Enterprise delivery orgs and InfoBeans' own sales team	The demo is InsaneSDD's stated lead-gen tool (per InfoBeans investor deck); a visual preview materially strengthens every sales demo	Investor Presentation, Mar 2026

3. Problem solved

The blind approval gate. Stakeholders must approve a UX specification they cannot visualize.
Consequences:

1. Non-technical approvers rubber-stamp text they can't evaluate, hollowing out the gate.
2. Design misunderstandings surface *after* automated code generation — the most expensive point in InsaneSDD's own pipeline.
3. The only recourse shown in the demo is "Re-run UX Agent" or "Request Revision" for the **entire** design — no way to approve four screens and flag one.

Evidence the gap is real (state as fact, not assumption):

- The OG demo shows the UX module output exclusively as text/lists (3 personas, 3 journeys, 5 screens, 30 components, design system) with no rendered screen.
- insanesdd.ai makes no mention of visual prototyping.
- Expona (InfoBeans' own RAG-grounded agent, citing internal `Insane_SDD.pdf`) confirmed: InsaneSDD "lacks automated UI prototyping" and "manual UI mockups slow iteration," and ranked a Spec-to-UI module as the #1 complementary module for a Frontend Lead.

4. Relationship to InsaneSDD

- **Complement, not critique.** Slots between the existing "Generated UX Design" review screen and the Agile Editor. Strengthens the existing approval gate; replaces nothing.
- **Input contract:** consumes the same UX specification structure the UX Design Agent already produces (personas, journeys, screens, components, design system, navigation, accessibility requirements) — modeled as a typed `UXSpec` JSON document.
- **Output contract:** emits per-screen approval states and structured revision requests linked to the frozen baseline version (v1), feeding InsaneSDD's traceability/audit story.
- **Defensive positioning:** if the live product renders any visuals not shown publicly, the module is presented as "deepening UX review into a per-screen approval workbench" — the framing holds either way.
- **Naming in all materials:** "a concept extension for InsaneSDD 2.0," never an official feature.

5. Complete user workflow

1. **Entry** — user opens UX Design Studio from the project (AgentPilot) after the UX Design Agent has produced a spec against frozen baseline v1.
2. **Overview** — dashboard summarizes the spec: 3 personas, 3 journeys, 5 screens, 30 components, design system, approval progress (e.g. "2 of 5 screens approved").
3. **Screen preview** — user selects a screen (Dashboard, Task Detail, Workflow Templates, Reports Export, Login); the rendering engine composes it live from the component registry, themed by the spec's design tokens.
4. **Interactive navigation** — clicking nav elements moves between screens per the spec's navigation design (desktop sidebar / mobile hamburger), so journeys can be walked end to end.
5. **Persona lens** — switch to "view as Taylor (Compliance Officer)": the screen is annotated with that persona's goals, frustrations, and journey touchpoints relevant to what's on screen.
6. **Responsive preview** — toggle mobile / tablet / desktop breakpoints; layout re-composes per the spec's mobile-first responsive strategy.
7. **Accessibility overlay** — toggle reveals WCAG 2.1 AA annotations on components: contrast check, ARIA roles/labels, screen-reader notes for live regions.

8. **Decision per screen** — **Approve** (timestamped, attributed, audit-logged) or **Request Revision** (structured form: affected component(s), issue category, description; logged against baseline v1).
9. **Regenerate screen** (*extension point*) — button invokes `DesignAgentProvider.regenerateScreen(screenId, revisionNotes)`; POC ships a mock provider returning a plausible variant, architected for a real LLM later.
10. **Gate completion** — when all screens are approved, the module surfaces "UX Design fully approved — ready for Agile plan generation," mirroring InsaneSDD's handoff to the Agile Editor.

6. Core product capabilities

1. Spec-driven **screen rendering engine** — composes React screens at runtime from `UXSpec` JSON via a component registry (~12–15 registered component types covering layout, forms, navigation, data display, feedback, charts).
2. **Design-token theming** — palette, typography hierarchy, and spacing scale from the spec applied at runtime (CSS custom properties); changing tokens re-skins every screen live.
3. **Interactive multi-screen navigation** honoring the spec's nav architecture.
4. **Persona lens** with contextual annotations for the three demo personas.
5. **Responsive breakpoint simulation** (mobile / tablet / desktop).
6. **Accessibility annotation overlay** (contrast, ARIA, screen-reader notes).
7. **Per-screen approval workflow** with structured revision requests and version linkage to the frozen baseline.
8. **Audit log view** of all approval/revision events.
9. **Provider-abstracted AI regeneration** (mocked in POC).

7. Frontend capabilities demonstrated

Skills evidence for the Lead Frontend evaluation, mapped to build artifacts:

- **Architecture:** typed domain model (`UXSpec`, `Screen`, `ComponentNode`, `DesignTokens`, `Persona`, `ApprovalEvent`); registry pattern for spec-to-component resolution; provider interface isolating AI from UI; documented state-management decision (approval/audit state vs. render state).
- **Rendering engineering:** recursive JSON-to-React composition, runtime theming via CSS custom properties, breakpoint simulation without page reload.
- **UX craft:** approval-workflow interaction design, empty/loading/error states, visual language matched to InsaneSDD's actual UI (light chrome, red accent `#E4002B`-family, sidebar layout — per the product screenshot).
- **Accessibility:** the studio itself meets WCAG 2.1 AA (keyboard navigation, focus management, ARIA) — practicing the standard the module inspects.
- **Performance:** lazy-loaded screen definitions, memoized rendering, debounced interactions — echoing the performance considerations in InsaneSDD's own generated UX spec.
- **Quality:** TypeScript strict; unit tests for the rendering engine and approval reducer; an Architecture Decision Record; deployed build (Vercel/Netlify) plus README.

8. Backend and AI integration requirements

POC (mocked): none — static `UXSpec` seed JSON mirroring the AgentPilot demo; approval state in memory/localStorage; `MockDesignAgentProvider` returns pre-authored screen variants with simulated latency.

Production path (documented in ADR, not built):

- GET /projects/:id/ux-spec (versioned), POST .../screens/:id/approval, POST .../screens/:id/revision-request, POST .../screens/:id/regenerate — the last invoking the UX Design Agent (demo used gpt-4.1-mini) with revision notes as context.
- Persistence for approval events (append-only), webhook/event emission into InsaneSDD's Live Terminal feed, and baseline-version linkage via the existing spec store.

9. Security and audit capabilities

- **Audit trail:** every approval, revision request, and regeneration is an append-only event {actor, action, screenId, baselineVersion, timestamp, payload} — consistent with InsaneSDD's "every AI decision, approval, and change request logged, traceable, and compliance-ready" positioning.
- **Traceability:** every event references the frozen baseline version (v1) and spec version, preserving requirement-to-design lineage.
- **RBAC (designed, mocked in POC):** roles Approver / Reviewer / Viewer; only Approvers can approve or request regeneration; POC includes a role switcher to demonstrate the model.
- **Versioning:** regenerated screens create a new screen version; prior versions remain viewable (POC: two pre-authored versions of one screen to demonstrate).
- **Production notes (ADR only):** SSO/OIDC alignment with the platform, signed audit events, retention policy for regulated industries (BFSI per InfoBeans' focus).

10. Presentation statement

"InsaneSDD's differentiator is that humans approve every gate before AI executes — but the UX gate asks stakeholders to approve a design they cannot see. The spec lists five screens and thirty components as text, and the first rendered UI appears only after code is merged. I validated this gap against your demo, your site, and Expona — your own agent confirmed InsaneSDD lacks UI prototyping and ranked this the top complementary module. So I built UX Design Studio: it renders the UX Design Agent's output as live, interactive, design-token-themed wireframes, lets each stakeholder review through their persona's lens, verifies accessibility inline, and turns approval into a per-screen, audit-logged decision tied to the frozen baseline. It strengthens your strongest feature — the approval gate — and because your demo is also your sales tool, it makes every sales conversation more convincing. It's built in React and TypeScript with the AI regeneration behind a provider interface, so plugging in the real UX Agent is an integration task, not a rewrite."

(~75 seconds spoken. Follow with the live demo.)

11. Exact scope summary

In scope (the 50 hours):

#	Deliverable	Hours
1	Typed UXSpec domain model + AgentPilot seed data (3 personas, 5 screens, ~30 components, design system, nav)	6

#	Deliverable	Hours
2	Rendering engine: component registry (~12-15 types), recursive composer, token theming	16
3	Per-screen approval + structured revision requests + audit log view + baseline linkage	8
4	Persona lens, journey walkthrough, responsive breakpoint preview	6
5	Polish: InsaneSDD visual language, a11y of the studio itself, loading/empty/error states, accessibility overlay	6
6	ADR + README + deployed build	4
7	Buffer + demo rehearsal	4
Total		50

Explicitly out of scope (say so in the README): drag-and-drop editing; real LLM calls (mock provider only); real backend/persistence beyond localStorage; real auth/SSO (role switcher only); multi-project/multi-repo; production-grade code-gen fidelity between wireframe and generated code; pixel-perfect visual design (wireframe fidelity is intentional — the artifact under review is structure, flow, and system, not final pixels).

Cut-line if hours run short (drop in this order): journey walkthrough → screen version history → accessibility overlay (keep at minimum contrast badges) → tablet breakpoint (keep mobile/desktop). Never cut: rendering engine, token theming, per-screen approval, audit log, seed-data fidelity to the demo.

Two usage notes. First, treat section 11's cut-line as binding when you're mid-build and tempted to add features — the document exists precisely so July-16-you doesn't renegotiate scope with July-13-you. Second, bring this document (or a one-page condensation of sections 1, 3, 10) to the interview itself; a frozen source-of-truth spec driving a build is *literally InsaneSDD's own methodology*, and pointing that out — "I ran your process on my POC" — is a closing line worth having.